



A marine coating: Self-healing, stable release of Cu^{2+} , anti-biofouling

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ABSTRACT

We developed a marine coating consisting of Cu-MOF-74, multi-walled carbon nanotube containing carboxyl groups (MWCNT-COOH) and self-healing polymers, which simultaneously possesses self-healing and anti-biofouling properties. Cu-MOF-74 can stably release Cu^{2+} by virtue of the coordination dissociative mechanism. Studies have proved that MWCNT can inhibit the growth of bacteria, so adding the MWCNT can help to reduce the amount of the copper ions and ensure the antibacterial effect of the coating. In addition, the cross-linked network and abundant -COOH provided by the polymers and MWCNT-COOH further prevent the loss of copper ions. Moreover, the coating we prepared has good performance of self-healing at room temperature or slightly heated because the polymers possess abundant non-covalent hydrogen bonds. Finally, the coating not only has superior antibacterial property, but also effectively prevents the adhesion of macrofouling organism. Therefore, the coating has a longer service life and its environmental friendliness has also been improved.

1. Introduction

Biological fouling (biofouling), resulted from those unwanted biological matter sticking to surfaces, including biofilms and macrofouling, can cause significant financial losses, environmental impact and health risks in the marine, medical and industrial fields (Bixler and Bhushan, 2012; He et al., 2021). Take marine biofouling for example, lots of underwater facilities, such as ships, buoys, sonar devices, pontoons, offshore structures, oil installations, platforms, underwater cables, underwater acoustic instruments, seawater cooling systems and marinas, are easy to be colonized by biofouling (Bixler and Bhushan, 2012). It can increase hydrodynamic drag, lower ship maneuverability, increase the emission of greenhouse gas, destroy port and fishery infrastructures and cause heavy economic burden (e.g., the cost of the US Navy on biofouling up to \$1 billion per annum.) (Rasmussen and Østgaard, 2001; Callow and Callow, 2002; Schultz et al., 2011; Lejars et al., 2012; Salta et al., 2013; Amini et al., 2017; Selim et al., 2020). To prevent biofouling, strategies including non-coating (electrical, irradiation, sonochemical techniques, surface cleaning. etc.) and coating techniques (biocidal antifouling paints, fouling-release coatings and fouling-resistant coatings) were developed (Selim et al., 2020). And the coating is the most convenient and effective technique among those strategies.

The coating based on biocides or enzymes is the earliest technique to prevent marine biofouling (He et al., 2021; Lejars et al., 2012). Since the prohibition of tin-based coatings (most powerful but highly poisonous), copper-based coatings have become the predominant antifouling biocide again (Dafforn et al., 2011). However, there are restrictions due to their toxic effects, release rates and tolerance of many fouling strains to such coatings (Selim et al., 2020). That is, stabilizing the sustained release of copper ions and formulating with additional other biocides will contribute to the development of copper-based coatings (Li et al., 2022; Chambers et al., 2006). In theory, metal ions chelated with ligands will be slowly released due to hydrolysis. Camilla Fant et al. (2006) prepared a PVM- M^{2+} -Medetomidine complex from poly(1-vinylimidazole-co-methyl methacrylate) (PVM), Cu^{2+} and Medetomidine. And they proved that the Medetomidine was more stable in this complex because of chelation. Therefore, we can also achieve the stable and sustained release of copper ions by chelating.

Metal organic frameworks (MOFs) are porous crystalline coordination polymers, which can be constructed from a wide range of metal ions and organic linkers (Liu et al., 2014; Dhakshinamoorthy et al., 2019; Veisi et al., 2021; Jun et al., 2022). They have been applied in many fields, such as catalysis, gas storage, dynamic materials and luminescent MOFs (Rosi et al., 2003; Bavykina et al., 2020; Perego et al., 2020; Perego et al., 2022). MOF-74 (also called CPO-27, $\text{M}_2(\text{dobdc})$ or

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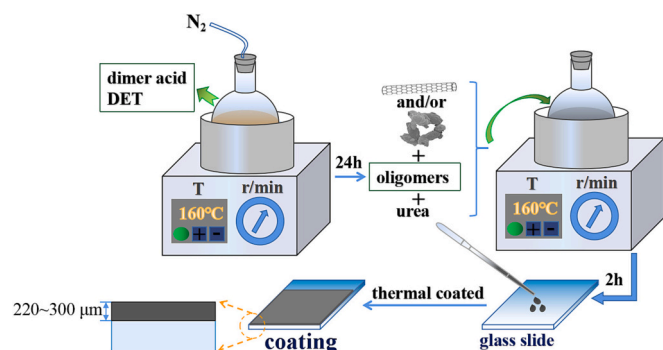


Fig. 1. The preparation process of the coating.

M₂(dhtp)) is a very unique and widely studied one among all MOFs. It consists of metal nodes (M) (M = Cu²⁺, Zn²⁺, Mg²⁺, Co²⁺, Cd²⁺, Mn²⁺, Fe²⁺ and Ni²⁺) and the organic ligand dobdc⁴⁻ (2,5-dioxido-1,4-benzenedicarboxylate) or dhtp⁴⁻ (2,5-dihydroxyterephthalate), which possesses good hydrolysis stability and highly stable structure (Liao et al., 2017; Xiao and Liu, 2019; Ling et al., 2022; Flores et al., 2021). From an antibiofouling point of view, Cu-MOF-74 is ideal for biological antifouling because of the existence of Cu²⁺ and decent stability against humidity. Fan et al. (2019) synthesized Cu-MOF-74 by a simple hydrothermal method and studied its inhibitory effect on the growth of *microcystis aeruginosa*. Zheng et al. (2021) used a facile and green method to synthesize Cu-MOF-74 to modify polyvinylidene fluoride (PVDF) membrane surface for improving anti-biofouling performance. The results they both obtained showed that Cu²⁺ released from Cu-MOF-74 as well as the reactive oxygen species (ROS) such as hydroxyl radicals ([•]OH) and superoxide radicals ([•]O²⁻) played important roles in the inhibition of bacterial and algal growth.

Carbon nanotubes (CNTs) classified as single-walled CNT (single layer, SWCNT) and multi-walled CNT (concentric multiple layers, MWCNT) are made of one-atom-thick carbon walls (graphene) rolling into long and hollow nanostructures (Dong and Ma, 2015; Thamaraiselvan et al., 2020; Yang et al., 2017). CNTs have extraordinary physiochemical properties such as mechanical stability, high surface area, high electrical and thermal conductivity, adsorption capacity and elasticity (Thamaraiselvan et al., 2020; Nam et al., 2015; Wu et al., 2021). In addition, most of the literatures have reported that CNTs have good antibacterial activity owing to the CNT size structure (smaller nanotube diameter facilitating the partitioning and partial penetration of nanotubes into the cell wall, a larger surface area for contact and interaction with the cell surface, unique chemical and electronic properties conveying greater chemical reactivity.) and additional attached groups (e.g., -OH and -COOH) (Kang et al., 2007; Kang et al., 2008; Katerina, 2010; Thamaraiselvan et al., 2018; Yang et al., 2010). Moreover, the CNT can also decrease the release rate of metal ions. Hao et al. (2016) prepared the MWCNTs-grafted carboxymethylated chitosan (CmCs) composites. And their results showed that the release rate of metal ions of the MWCNT-CmCs-metal was slower than CmCs-metal due to the existence of MWCNT. Therefore, formulating with MWCNT is a good choice to improve the property of copper-based coatings.

In this article, we have developed a Cu-MOF-74/MWCNT-COOH composite coating (Cu-CNT coating) and demonstrated its excellent anti-biofouling property. The coating was prepared using Cu-MOF-74, MWCNT-COOH and self-healing film-forming polymer (Fig. 1). Owing to chelation between metal ions and ligands in Cu-MOF-74 and the existence of abundant -COOH provided by the polymers and MWCNT-COOH, the release of Cu²⁺ in the coating is stable and sustained, and the amount of Cu²⁺ added is also reduced, which makes the coating long-term effectiveness. In addition, the coating exhibits excellent antibacterial effects, resistance to macrofouling organisms, and self-healing ability. These properties make the coating promising for

application in the field of marine antifouling.

2. Materials and methods

2.1. Materials

2,5-dihydroxyterephthalic acid (DHTP, 98 %), urea (99 %), methanol (99.5 %), ethanol (AR), yeast extract (AR), sodium chloride (AR), tryptone (AR), agar (AR), sodium sulfate decahydrate (AR), magnesium chloride hexahydrate (AR), calcium chloride anhydrous (AR) and glutaraldehyde (AR) were purchased from Aladdin Reagent Co., Ltd. Copper(II) acetate (Cu(OAc)₂, 98 %), diethylenetriamine (DET, 99 %) and multi-walled carbon nanotube containing carboxyl groups (MWCNT-COOH, > 95 %, inner diameter: 5–10 nm, outer diameter: 20–30 nm, length: 10–30 μm, -COOH: ~1.2 wt%) were provided by Macklin Reagent Co., Ltd. Dimer acid (80 %) was purchased from Yuan Ye Biotechnology Co., Ltd. Chloroform (AR), acetone (AR) and nitric acid (GR) were provided by Guangzhou Chemical Reagent Factory. Artificial seawater (ASW) was fabricated as GB/T 7790-1996. Copper standard solution (1000 ppm) was obtained from Guangzhou Testing Technology Co., Ltd. Deionized water (DI water) and *Escherichia coli* (*E. coli*, DH5α, DingGuo) were provided by local laboratory. No additional processing was done for the materials.

2.2. Preparation of Cu-MOF-74

The Cu-MOF-74 was synthesized by a facile and green method. Firstly, 0.4 g of Cu(OAc)₂ was dissolved in 10 mL of methanol. Then another 5 mL of methanol was used to dissolve 0.2 g DHTP. After that, the DHTP solution was added dropwise into the Cu(OAc)₂ solution. The mixture was magnetically stirred for 24 h at room temperature to obtain a kind of reddish-brown crystalline solid containing Cu-MOF-74. The solid was collected by centrifugation and then immersed in 20 mL MeOH for 24 h and 4 times to fully wash away impurities. At last, the resulting Cu-MOF-74 was collected by centrifugation and dried in an 85 °C vacuum drying oven for 5 h.

2.3. Preparation of oligomers

Dimer acid is usually referring to dimeric fatty acid. The main component of dimer acid contains two carboxylic acid groups. Here it was reacted with diethylenediamine (DET) to form oligomers, which can enhance the degree of crosslinking and is helpful to form hydrogen bond because of the introduction of N atom. Specifically, 8.3 g of dimer acid and 3.4 g of DET were mixed and reacted at 160 °C for 24 h under stirring and a N₂ atmosphere. Then, the crude product was dissolved in 20 mL chloroform and transferred to the separating funnel with 40 mL water and 20 mL methanol. After shaking and standing to separate, the lower solution in the separating funnel was collected. At last, the solution was put into a vacuum oven for 24 h to obtain the oligomer.

2.4. Preparation of Cu-CNT coating

2.2 g of the as-prepared oligomer was mixed with Cu-MOF-74 and MWCNT-COOH to obtain a homogeneous suspension. Then, urea (0.5 g) was added. The above mixture reacted for 2 h at 160 °C with constant mechanical stirring to form self-healing materials. At last, self-healing materials were thermal coated on glass slides (2.5 cm × 2.6 cm, glass slides were respectively sonicated in acetone and aqueous solution for 10 min and then transferred to absolute ethanol before use). After cooling, the hybrid coatings ('coatings' refer to glass slides coated with self-healing materials) were obtained (The thickness of the material coated on the glass is 220–300 μm, Figs. 1 and S3). It should be noted that in order to avoid the potential impact of chemical substances generated by metal substrate corrosion on antibacterial experiments, glass slides were selected for coating preparation. The contents of Cu-

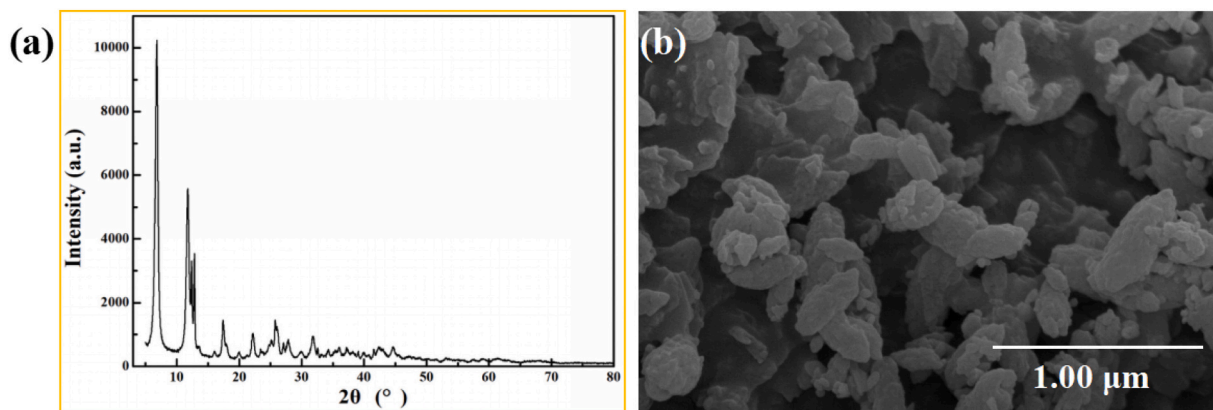


Fig. 2. (a) XRD pattern and (b) SEM micrograph of Cu-MOF-74.

MOF-74 and MWCNT-COOH in different coatings are showed in Table S1.

2.5. Characterizations

The morphology of Cu-MOF-74 was confirmed by scanning electron microscopy (SEM, regulus 8230). The crystal structure of Cu-MOF-74 was characterized by the X-ray diffraction technique (XRD, Empyrean). Thermal stability of the Cu-MOF-74 was evaluated by thermogravimetric analysis (TG, Netzsch, TG209F1 libra, N₂ flow, 28–510 °C, 2 °C min⁻¹). Field emission transmission electron microscope (TEM, FEI Talos F200X) was selected to observe the distribution of Cu-MOF-74 and MWCNT-COOH in the coatings. A field emission scanning electron microscope (FE-SEM, Gemini300) was applied to measure the thickness of the coating. The element mapping of the coating before and after immersion in ASW was detected by an energy-dispersive spectrometer (EDS, Oxford SYMMETRY).

2.6. Self-healing assay

This experiment was conducted using the methods of Hui Yang et al. (2016). A cross-shape pattern was carved on the surface of the coating by a knife. And then all of them were placed on an experiment table at room temperature for 3 h. During this period, it should be continuously observed whether the cross shaped pattern on the coating disappears. After 3 h, the coatings whose cross shape did not disappear were put into a 90 °C oven for 10 min. After that, the coatings were took out and observed the presence of cross patterns on the coatings again.

2.7. Cu²⁺ leaching trial

The coatings were immersed in ASW in a big crystal dish ($r \approx 176$ mm, room temperature) under magnetic stirring (630 rpm/min) for 1 day. And then every coating was immersed in 50 mL of fresh ASW under magnetic stirring (550–650 rpm/min) for 10–11 h to obtain leaching solution. And then the leaching solution was diluted ($\rho_{\text{salt}} \leq 500 \text{ mg L}^{-1}$) by 0.5 % nitric acid to detect (atomic absorption spectrometer, Thermo, SOLAAR ICE 3500) and record the metal ions in the coating for 1 day exudation concentration (Copper ion release rate = $\rho(\text{Cu}) \times V \times 24 / (t \times A)$). $\rho(\text{Cu})$: concentration of copper ion, $\mu\text{g/L}$; V : volume of ASW, L; A : area of the coating, cm^2 ; t : time for coating impregnation, h). Afterwards we replace the remaining solution in the big crystal dish with fresh ASW and use the same method to detect the release concentration of metal ions after the coatings were immersed for 3, 7, 10, 14 and 21 days, respectively.

2.8. Hydroxyl radicals ($\cdot\text{OH}$) and superoxide radicals ($\cdot\text{O}_2^-$) assays

Electron paramagnetic resonance instrument (EPR, Bruker A300–10/12) was used to test whether the coatings (the control, MWCNT-II, Cu-MOF-74-II and Cu-CNT-II coatings were immersed in 0.9 % NaCl solution for 24 h before testing) can promote the generation of $\cdot\text{OH}$ and $\cdot\text{O}_2^-$.

2.9. Antibacterial assays

The colonization of bacteria can form a biofilm, which refers to ‘microfouling’ or ‘slime’. Here, we used *Escherichia coli* (*E. coli*) as the model bacteria to study the antibacterial performance of the coating.

We precultured bacteria (*E. coli*) by dipping the bacterial solution (the bacterial solution used for precultivation was taken from the local laboratory) and scribing on the solid culture medium, then incubating at 37 °C for 24 h. Then the precultured *E. coli* (1 CFU, CFU = colony forming units) was transferred into 10 mL of Luria-Bertani (LB) medium, cultured for 8 h (10^7 – 10^8 CFU mL⁻¹) at 37 °C, 180 rpm. After that, the solution was centrifuged to pellet and resuspend the bacteria in 10 mL of 0.9 % NaCl solution. Next, 300 μL of the bacterial suspension was diluted 100 times (10^5 – 10^6 CFU mL⁻¹) with 0.9 % NaCl in a 50 mL sterile centrifuge tube, and then coating (immersing in ASW 1, 4, 10, 17 days respectively before use) was added. The centrifuge tubes were put in the same shaker for 12 h (37 °C, 180 rpm). After that, the centrifuge tubes were taken out and shaken vigorously (all the bacteria are distributed in the solution as evenly as possible). Then, 1 mL of bacterial solution was transferred from different centrifuge tubes and gradient diluted (dilution factor is recorded as 10 $\times$, 100 $\times$ and 1000 $\times$). The number of surviving colonies were counted by the plate counting method (100 μL) and recorded as the number of surviving bacterial colonies after 18–24 h. The experiments were repeated at least two times. The calculation method for antibacterial rate is as follows: Bacteriostasis rate = $(1 - \text{Ne}/\text{Nc}) \times 100\%$, Ne: number of colonies of the experimental group, Nc: number of colonies of the control group.

2.10. Morphology of *E. coli*

1 CFU of *E. coli* was transferred into 10 mL of LB medium, cultured for 24 h at 37 °C, 180 rpm. The prepared *E. coli* were resuspended with 0.9 % NaCl after centrifugation, and then divided into four equal parts and put into the 50 mL sterile centrifuge tube respectively. Then the coatings (the control, MWCNT-II, Cu-MOF-74-II and Cu-CNT-II coatings) were put into these sterile centrifuge tubes in which 0.9 % NaCl solution was added until the coatings were submerged. After shaking 24 h at 37 °C, 180 rpm, bacteria were separated and washed with 0.9 % NaCl solution and fixed with 3 mL of 2.5 % glutaraldehyde at 4 °C for 24 h. After that, the bacteria were washed again with 0.9 % NaCl and

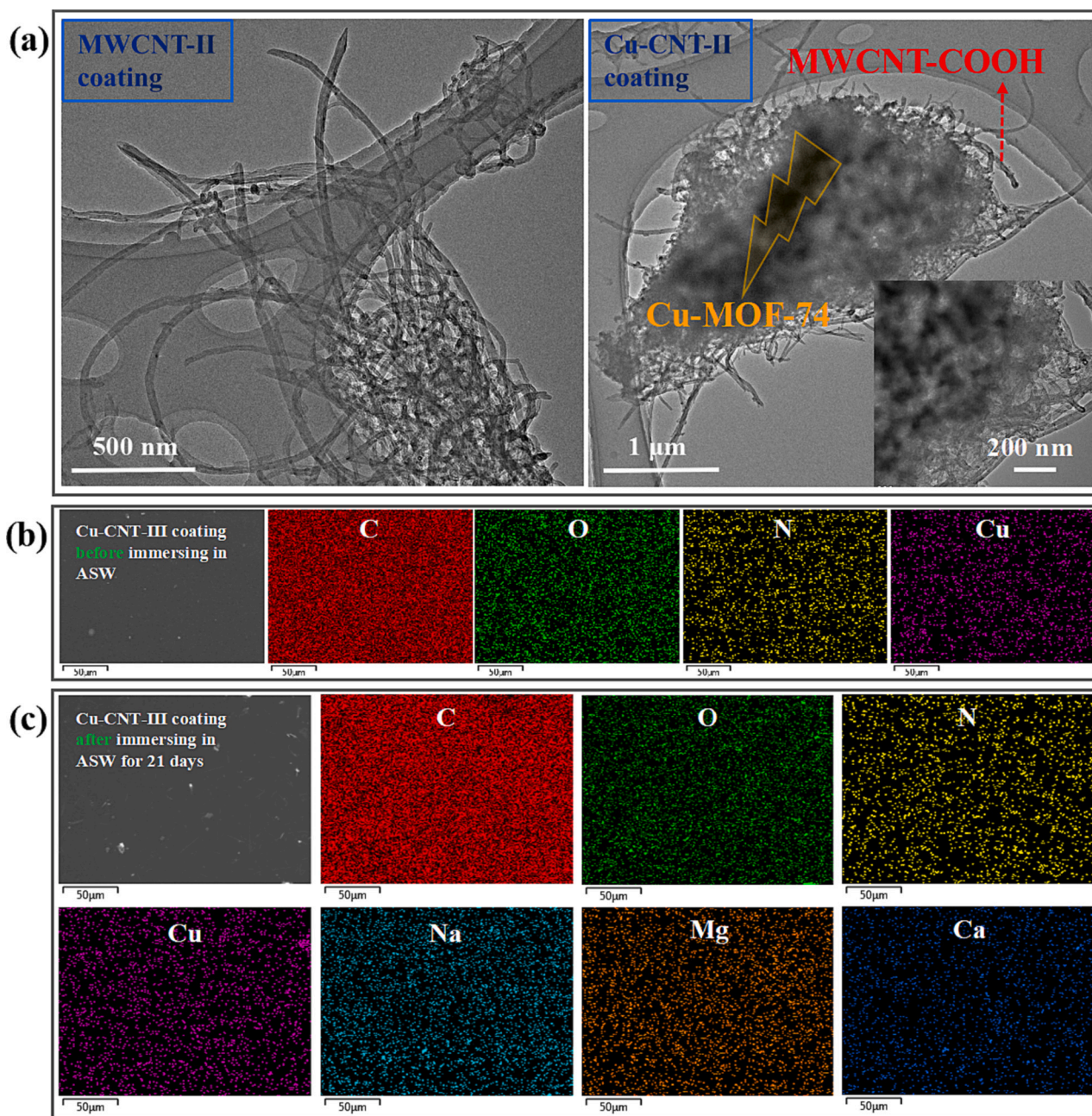


Fig. 3. (a) TEM micrograph of MWCNT-II and Cu-CNT-II coating. (b) Mapping images of Cu-CNT-III coating before immersing in ASW. (c) Mapping images of Cu-CNT-III coating after immersing in ASW.

dispersed in 50 %, 75 % and 100 % ethanol in sequence for dehydration (15 min). Finally, they were resuspended in absolute ethanol for later use. The SEM (regulus 8230) was used to observe the morphology of the bacteria.

2.11. Real sea experiment

The coatings were fixed on plastic mesh and then immersed in the local sea about two meters deep for one month. After retrieving the coatings from the sea, the growth of organisms on them was observed. Then the performance of the coatings against macrofouling organisms (e.g., barnacle) can be evaluated.

3. Results and discussion

3.1. Characterization of Cu-MOF-74

As shown in the XRD pattern, there are two high and sharp diffraction peaks at $2\theta = 6.8^\circ$ and 11.8° (Fig. 2a), and these peaks are consistent with the characteristic diffraction peaks of Cu-MOF-74 reported in many literature (Fan et al., 2019; Flores et al., 2018; Sanz et al., 2013). The result indicates that the Cu-MOF-74 was successfully synthesized. Furthermore, the SEM micrograph shows that the particles in the sample of Cu-MOF-74 are composed by agglomerates/aggregates formed by a large number of nanocrystals and their size is well below micrometer scale (Fig. 2b). In addition, the TG curve shows that Cu-MOF-74 will not decompose below 200°C (Fig. S2), indicating that coating prepared at 160°C will not have adverse effects on Cu-MOF-74.

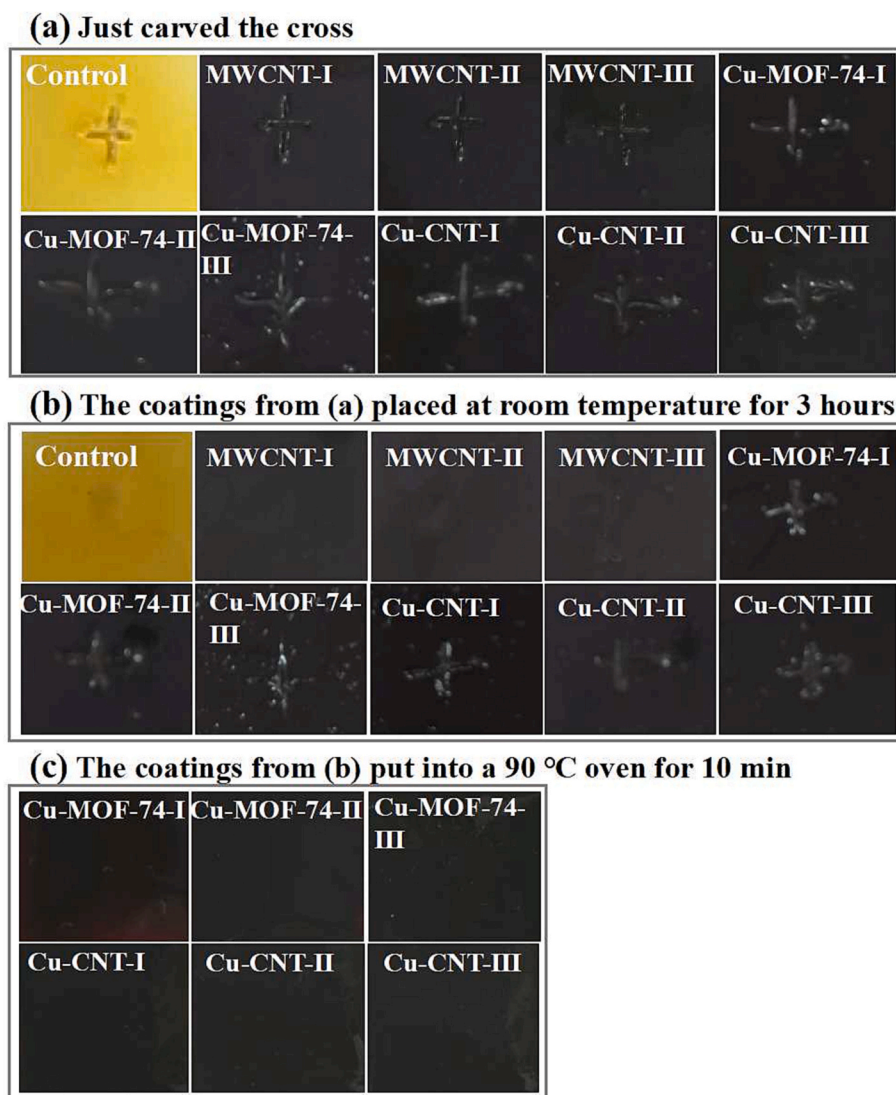


Fig. 4. (a) Cross-shape patterns carved on the surface of the coatings by a knife. (b) The coatings from (a) stood still at room temperature for 3 h and (c) six coatings from (b) put into a 90 °C oven for 10 min. After two steps of (b) and (c) treatment, the cross on the coating disappears and all coating surfaces were restored to flatness.

3.2. Characterization of coatings

Here the distribution of MWCNT-COOH and Cu-MOF-74 in coatings are observed. As can be seen from Fig. 3a, slender carbon nanotubes are observed in MWCNT-II coating, and these carbon nanotubes are intertwined with each other. Moreover, it can be observed that carbon nanotubes are entangled with Cu-MOF-74 particles in Cu-CNT-II coating. Furthermore, the distribution of elements on the coating surface was observed through mapping. As shown in Fig. 3b, before immersion in ASW, C, O, N, and Cu elements are distributed on the coating surface, with Cu accounting for 2.3 wt% (Table S2). After soaking the coating in seawater for 21 days (Fig. 3c and Table S3), the surface is distributed with C, O, N, Cu (2.81 wt%), Na (0.78 wt%), Mg (0.10 wt%) and Ca (0.14 wt%). There is no significant changes in Cu content before and after immersion, indicating that the coating can stably and continuously provide Cu^{2+} . In addition, Na, Mg and Ca are detected on the surface of the coating after immersion, but their contents are low, indicating that the coating can adsorb a small amount of metal ions in ASW.

Fig. S4 shows the FTIR spectrogram of the prepared coatings. It can be seen from the Control (only polymers coated on glass slide) one that

two peaks in the range 2750–3000 cm^{-1} attribute to the stretching vibration of C=C in polymers. A characteristic absorption peak at 1549 cm^{-1} belongs to the characteristic absorption peak of C=O in amide groups (produced by the reaction of the carboxyl group of dimeric acid with diethylenetriamine). The absorption peak at 3285 cm^{-1} can be associated with the O—H stretching vibration. The MWCNT-II curve has additional peaks at 759 and 1499 cm^{-1} compared to the Control, and the curve of the coatings containing Cu-MOF-74 (Cu-MOF-74-II and Cu-CNT-II) have one more peak at 1603 cm^{-1} compared to the MWCNT-II, indicating that MWCNT and Cu-MOF-74 are mixed with the polymers. Importantly, the peaks at 1549 and 3285 cm^{-1} in the Cu-MOF-74-II and Cu-CNT-II curves are weaker than that in both the Control and MWCNT-II curves, implying that Cu^{2+} have reacted with -COOH.

3.3. Self-healing property

The cross-shape patterns on the coatings of Control, MWCNT-I, MWCNT-II and MWCNT-III disappeared within 3 h at room temperature, indicating these coatings have rapid self-healing properties (Fig. 4b). However, the coatings containing Cu-MOF-74 (including Cu-MOF-74-I, Cu-MOF-74-II, Cu-MOF-74-III, Cu-CNT-I, Cu-CNT-II and Cu-

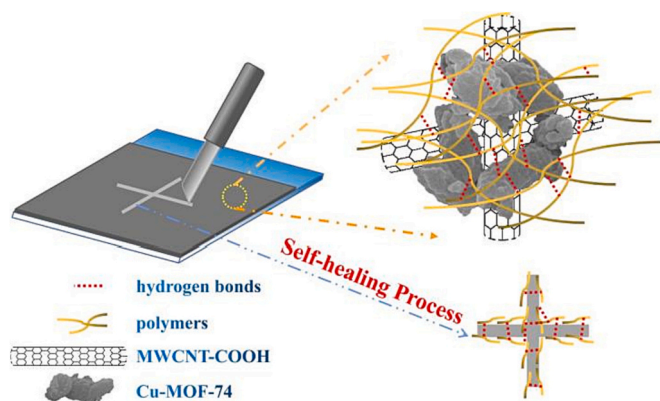


Fig. 5. Schematic illustration of those coatings based on hydrogen bonds and the self-healing process of coatings.

CNT-III coating) could not repair themselves after placing at room temperature for 3 h. But they were able to heal after being heated at 90 °C for 10 min (Fig. 4c). The self-healing polymers possess abundant non-covalent hydrogen bonds, which endows the coating with the ability to repair itself repeatedly in a short time (Yang et al., 2016). In addition, due to the presence of -COOH and its small diameter in MWCNT-COOH, the self-healing effect of the coatings containing MWCNT-COOH (without Cu-MOF-74) is also significant. However, due to the reduction in hydrogen bond formation of polymer chains via hindering the contact between polymer chains by the large particles of Cu-MOF-74 (Fig. 2b), the self-healing performance of the coating deteriorates. Nevertheless, after heating, the segments of polymers disentangle from MWCNT and Cu-MOF-74 (Yang et al., 2016), then the hydrogen bonds can be formed again to repair themselves (Fig. 5).

3.4. The release rate of Cu^{2+}

As we know, copper is one of the essential trace elements in animal and plant tissues and partakes in biological processes (Cady et al., 2011; Marelli et al., 2015). However, excessive discharge of Cu^{2+} will do harm to environment. Therefore, reducing the release of copper ions can not only extend the service life of the coating, but also alleviate the potential harm of copper ions to the environment.

Cu-MOF-74 can release Cu^{2+} in solution (Fan et al., 2019). It can be seen from Fig. 6a, the concentration of Cu^{2+} released from Cu-MOF-74 coatings (Cu-MOF-74-I, Cu-MOF-74-II, Cu-MOF-74-III) ascends with the increase of Cu-MOF-74 content after being immersed in ASW for one

day. From day 3 on, the release rate of Cu^{2+} begins to stabilize, including all of the coatings (Cu-MOF-74-I, Cu-MOF-74-II, Cu-MOF-74-III, Cu-CNT-I, Cu-CNT-II, Cu-CNT-III), and the release rate of Cu-MOF-74-III is only 1.3 times of that of Cu-MOF-74-II on day 3 (1.8 times on day 1). This is possibly because the Cu-MOF-74 on the surface was soaked out first on day 1. Afterwards, the Cu^{2+} inside the polymer dissolve out, but the cross-linked network and rich -COOH of polymer help resist the loss of Cu^{2+} . Among the Cu-CNT coatings, the concentration of Cu^{2+} released from Cu-CNT-I remains steady due to the small content of Cu-MOF-74. Interestingly, the concentration of Cu^{2+} from Cu-CNT-III coating is always less than that of the Cu-MOF-74-II and Cu-CNT-II coatings within 21 days. Among the three coatings, the content of Cu-MOF-74 in Cu-CNT-III coating is the same as that in Cu-MOF-74-II coating and half in Cu-CNT-II coating. This indicates that the MWCNT-COOH works on the decrease of Cu^{2+} release rate. And this is because as the MWCNT-COOH content increases, more -COOH groups are introduced to chelate with Cu^{2+} . Relatively speaking, this coating has a better ability to continuously and stably release Cu^{2+} , which is beneficial for the long-term use of the coating (Table S4).

3.5. The production of $\cdot\text{OH}$ and $\cdot\text{O}^{2-}$

Fig. 7 shows the signal diagram of the generated $\cdot\text{OH}$ and $\cdot\text{O}^{2-}$ by the coatings (the control, MWCNT-II, Cu-MOF-74-II and Cu-CNT-II coating). The MWCNT-II, Cu-MOF-74-II and Cu-CNT-II coating have typical $\cdot\text{OH}$ and $\cdot\text{O}^{2-}$ signal while the control group have no signal. The results indicate that the coatings containing Cu-MOF-74 and/or MWCNT-COOH are able to promote the generation of $\cdot\text{OH}$ and $\cdot\text{O}^{2-}$, thereby causing imbalance of $\cdot\text{OH}$ and $\cdot\text{O}^{2-}$ (belong to reactive oxygen species) in cells and damaging them. Besides, for the coatings of MWCNT-II, Cu-MOF-74-II and Cu-CNT-II, the signal intensity of their radicals (both $\cdot\text{OH}$ and $\cdot\text{O}^{2-}$) strengthens in sequence, implying that their ability to promote the production of free radicals also increases in turn.

Reactive oxygen species (ROS), such as hydroxyl radicals ($\cdot\text{OH}$), superoxide radicals ($\cdot\text{O}^{2-}$), singlet oxygen ($^1\text{O}_2$), hydrogen peroxide (H_2O_2) and hypochlorous acid (HOCl), are in a dynamic equilibrium in normal cells, and can act as intracellular signaling molecules (Strzelczyk and Wiczowski, 2012; Loprasert et al., 1996; Irani, 2000). Once the ROS level is imbalanced, it will damage cells by destroying organic components (e.g. proteins and DNA) (Zhou et al., 2014; Fu et al., 2015; Sun et al., 2022).

3.6. Evaluation of long-term antibacterial property

The long-term antibacterial property of the prepared coatings is

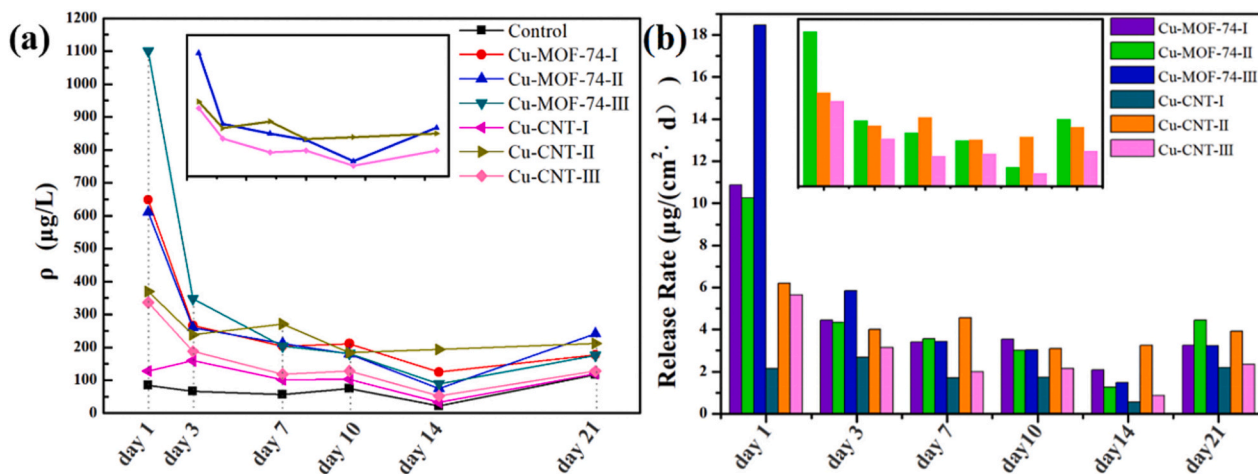


Fig. 6. (a) Concentration (ρ) of copper ions released from different coatings after immersed in the ASW for different days. (b) Copper ion release rate after immersed in the ASW for different days.

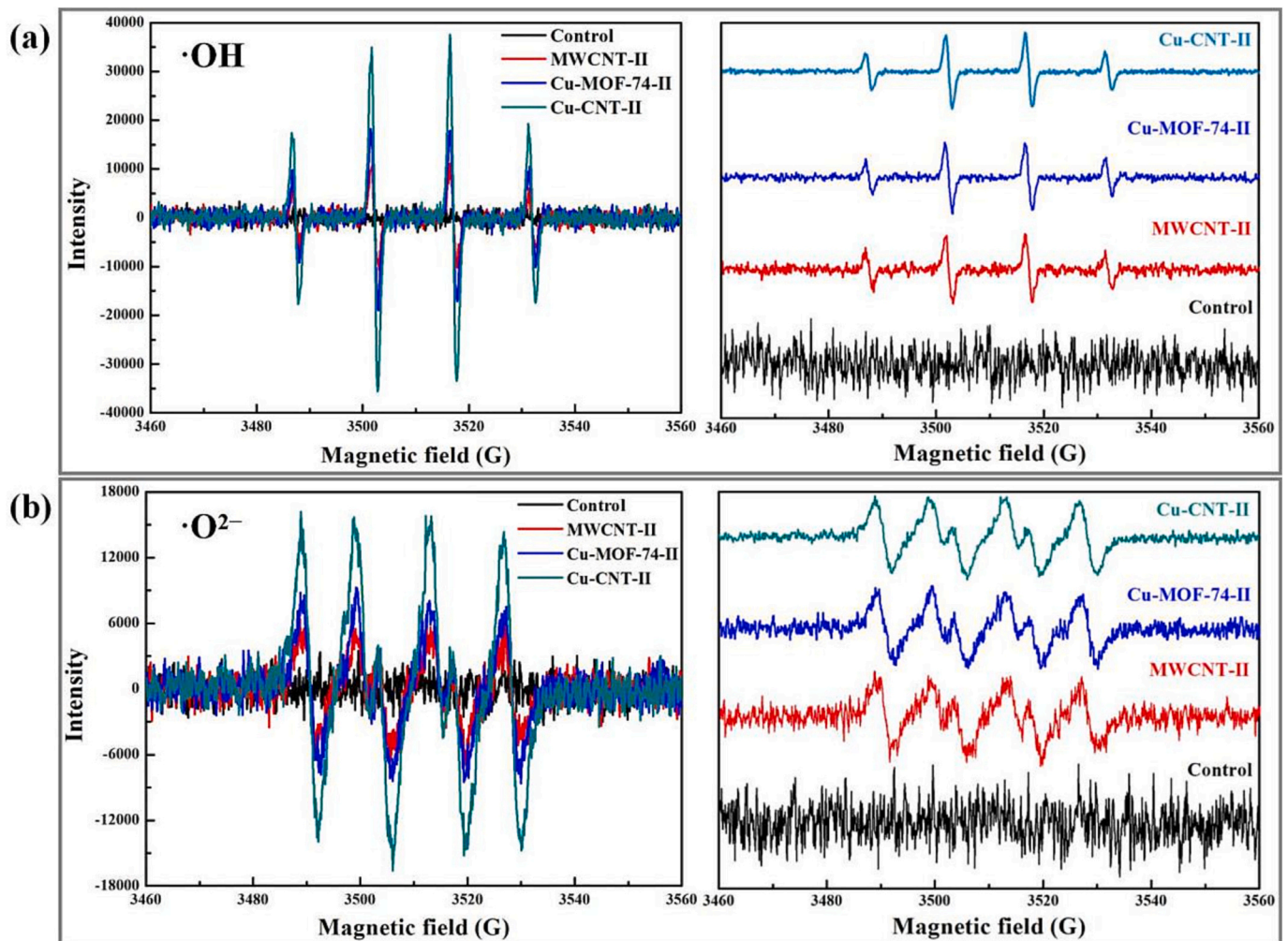


Fig. 7. The ability of coatings to promote the production of (a) $\cdot\text{OH}$, (b) $\cdot\text{O}_2^-$.

evaluated. The coatings, MWCNT-I and MWCNT-III, are poor to kill bacteria on day 1 (Fig. 8a). The former is because MWCNTs are added less and the soaking time is short, resulting in very few separated MWCNTs. The latter is attributed to the larger amount of MWCNTs added, resulting in tighter entanglement between nanotubes and difficulty in dissolution. However, their antibacterial rates have increased on day 4 (Fig. 8b, antibacterial rate of all experimental coatings: $\sim 100\%$). This is attributed to the increase in immersion time, resulting in the separation of more carbon nanotubes. Other coatings (MWCNT-II, Cu-MOF-74-I, Cu-MOF-74-II, Cu-MOF-74-III, Cu-CNT-I, Cu-CNT-II, Cu-CNT-III) show excellent antibacterial performance on day 1 and day 4. By the 10th day, the antibacterial property of the coatings is still good, and the antibacterial rates are $>83\%$ (Fig. 8c). On the 17th day, the data gives prominence to the continuous antibacterial effect of the composite coating, Cu-CNT-II, whose antibacterial rate is $>93\%$ (Fig. 8d). This is attributed to the coating being able to release more copper ions ($\sim 0.25 \text{ mg L}^{-1}$, Fig. 6a) while promoting more ROS production (Fig. 7). It is noted that the antibacterial rate of the Cu-MOF-74-I coating remained at a high level within 17 days, as the copper ion release of the Cu-MOF-74-I coating has always been relatively high. In addition, the antibacterial rates of the MWCNT coatings are extremely poor on day 17, even negative, probably because MWCNTs provide a carbon source for bacteria in the later stage, but instead promote their growth. These results demonstrate that the Cu-CNT-II coating have sustained good antibacterial performance and can stably release copper ions, which contributes to prolonging the service life of coating. And the antibacterial rate of the

coatings we prepared on the first day can almost reach 100% , better than most of the materials (the antibacterial rates of the membranes listed by Zheng et al. (2021) range from 32.3% to 99.4%).

CFU values are expressed as means $\pm$ standard error, $n \geq 2$. “ $\times 100$ ” represents the actual number of bacteria = the value of each column multiplied by 100 (Because the bacterial suspension was diluted 100 times before adding the coating). The antibacterial rate of the coating is calculated from the average value of CFU.

3.7. Morphology of *E. coli*

Fig. 9 shows the surface morphology of bacteria after interaction with several typical coatings. From the Fig. 9a, it can be seen that the *E. coli* cultured with the control coating maintain their complete shape. However, the *E. coli* from the experimental groups show incomplete concave appearance. This is because the bacterial cell membrane has been destroyed, leading to leakage of cell contents. Moreover, the longitudinal size of bacteria are $2.8 \mu\text{m} \sim 3.7 \mu\text{m}$ (Control), $1.9 \mu\text{m} \sim 2.7 \mu\text{m}$ (MWCNT-II), $1.7 \mu\text{m} \sim 2.1 \mu\text{m}$ (Cu-MOF-74-II) and $1.6 \mu\text{m} \sim 2.9 \mu\text{m}$ (Cu-CNT-II), respectively. The size of the last three bacteria is smaller than that of the control group. This suggests that the coatings containing Cu-MOF-74 and/or MWCNT-COOH have influence on the growth of bacteria. The above results indicate that when exposed to Cu-MOF-74 and MWCNT-COOH, bacterial growth is affected and cell walls can be damaged.

There are many mechanisms for antibacterial activity. Copper ions

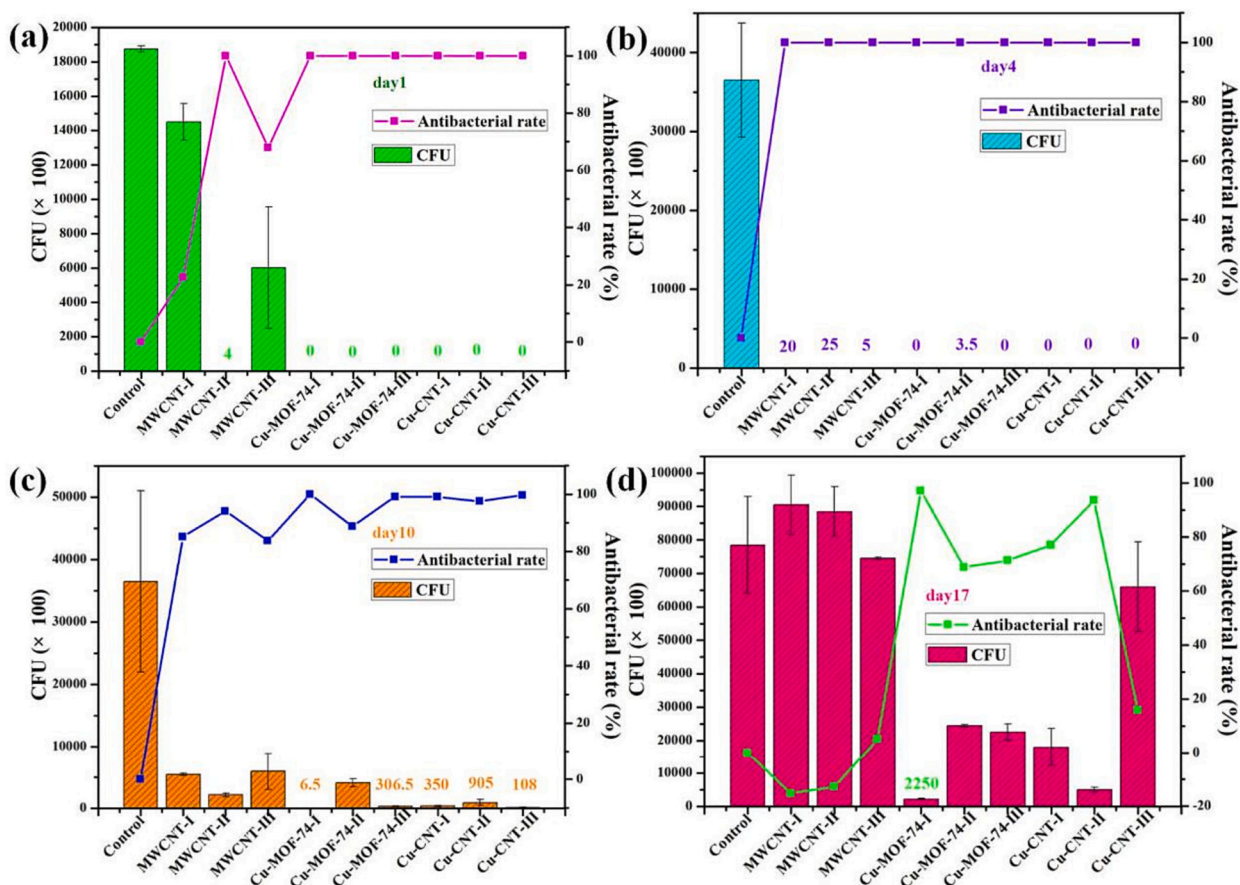


Fig. 8. The surviving bacterial colonies (CFU) and antibacterial rate curves of (a) coatings immersed in ASW for 1 day, (b) coatings immersed in ASW for 4 days, (c) coatings immersing in ASW for 10 days and (d) coatings immersing in ASW for 17 days. The.

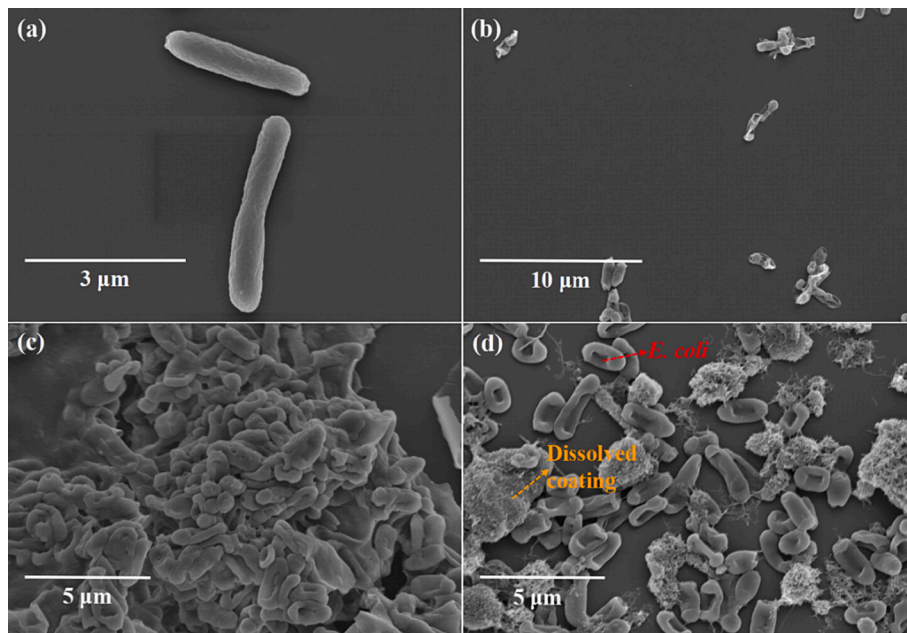


Fig. 9. The morphology of *E. coli* after cultured with different coatings. (a) Control, (b) MWCNT-II, (c) Cu-MOF-74-II, (d) Cu-CNT-II.

can react with the -NH₂, -SH and -COOH groups provided by protein molecules in the cell, which causes some macromolecules to be destroyed and then makes the organism to lose its biological function and inactivation. In addition, literature reported that single MWCNTs or

MWCNT aggregates may enter the cell membrane via direct penetration or endocytosis and then inactivate the cell (Yang et al., 2017; Zhu et al., 2016; Shi et al., 2011). Moreover, the functional group (such as -COOH, -OH) distributed on MWCNTs may affect cell membranes by interacting

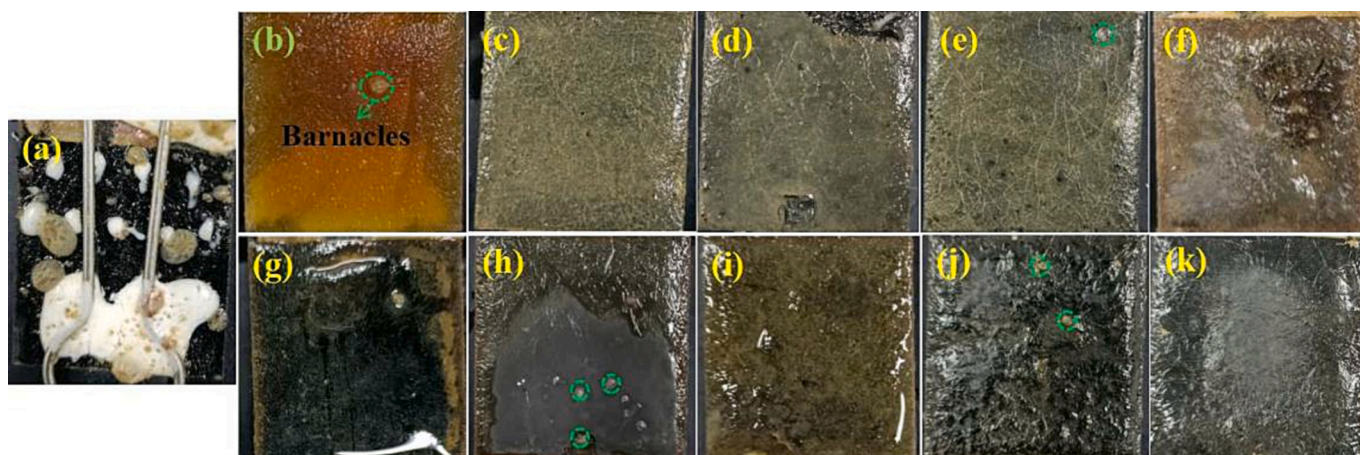


Fig. 10. Coatings after immersing in local shallow maritime space for one month. (a) Blank (consisted of glass, ironwork and silicone sealant, no polymers, Cu-MOF-74 and MWCNT-COOH), (b) Control (only polymers coated on glass slide), (c) MWCNT-I, (d) MWCNT-II, (e) MWCNT-III, (f) Cu-MOF-74-I, (g) Cu-MOF-74-II, (h) Cu-MOF-74-III, (i) Cu-CNT-I, (j) Cu-CNT-II, (k) Cu-CNT-III.

with bacterial membranes through a cross-linking reaction. Furthermore, MWCNT-COOH tightly adhering to and interacting with cell surface and altering cell permeability resulting can lead to the easier entrance of Cu^{2+} into cells to chelate with $-\text{NH}_2$, $-\text{SH}$ and $-\text{COOH}$ groups, then the cells are damaged. Finally, ROS such as $\cdot\text{OH}$ and $\cdot\text{O}^{2-}$ produced by Cu-MOF-74 and MWCNT-COOH can also cause oxidative damage to the cells (Fan et al., 2019; Zheng et al., 2021; Dong and Ma, 2015; Kang et al., 2008).

3.8. Antibiofouling property of the coatings in the local sea

There are >4000 species of fouling organisms which are primarily the attached or sessile forms occurring naturally in the shallower water along the coast (Yebra et al., 2004). The results of real sea experiment shows that the blank coating (consisted of glass, ironwork and silicone sealant, no polymers, Cu-MOF-74, MWCNT-COOH) cannot inhibit the settlement of fouling organisms (mainly barnacles here). But all coatings exhibit inhibition of macrofouling organisms settlement compared to the blank and control group (only polymers coated on glass slide) (Fig. 10). Although some settlements occur on the coatings (MWCNT-III, Cu-MOF-74-III, Cu-CNT-II), the growth of barnacles is clearly inhibited due to the size of the barnacles on them are smaller than those on the blank and control. In addition, it can be noted that more barnacles settle on the Cu-MOF-74-III and Cu-CNT-II coating than on the MWCNT-III coating, which is due to the increased hardness of the coating after the addition of Cu-MOF-74. Actually, in order for barnacle larvae to complete their transition to adulthood, their “Cyprus” form must adhere to a hard substrate (Callow and Callow, 2002). Most coatings (MWCNT-I, MWCNT-II, Cu-MOF-74-I, Cu-MOF-74-II, Cu-CNT-I, Cu-CNT-III) do not have barnacle fixation on the surfaces, indicating that the coatings has excellent resistance to the colonization of macrofouling organisms. That is to say, the coatings have good antifouling performance.

4. Conclusions

In conclusion, we have successfully prepared several coatings consisting of Cu-MOF-74, MWCNT-COOH and self-healing polymers. The copper ions leaching trial shows that the Cu^{2+} release from Cu-CNT coatings is more stable than that of Cu-MOF-74 coatings, and the release amount is lower at the same MOF content ($\rho_{\text{Cu-CNT-III}} < \rho_{\text{Cu-MOF-74-II}}$). In addition, the coatings have the ability to repair itself due to the abundant non-covalent hydrogen bonds in the polymers. These make the coatings have a long service life and alleviate the possible damage caused by copper ions to the environment. At last, it has been shown that all coatings containing Cu-MOF-74 and/or MWCNT-COOH are able to

kill bacteria with high antibacterial rate and possess long-term effectiveness on the resistance of macrofouling organisms by combining the advantages of Cu-MOF-74 and MWCNT-COOH. Among these coatings, the Cu-CNT-II coating has more stable antibacterial effect, and its antibacterial rate can reach 93 % at its lowest. Therefore, the coating is expected to be applied in the field of anti-biofouling.

CRediT authorship contribution statement

Liuqin Zhang: methodology, experiment, data processing, writing and editing. Huali Li: experiment and editing. Xiaohu Zhang: editing and supervision. Qiuping Li: supervision and investigation. Guangyu Zhu: investigation and validation. Fa-Qian Liu: funding acquisition, investigation, methodology.

Declaration of competing interest

The authors declare no conflict of interest.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.marpolbul.2023.115524>.

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